Antareep Kumar Sarma

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Education

École Polytechnique Fédérale de Lausanne — Lausanne, Switzerland
P.hD., Doctoral School in Mechanics — *Sept 2023 – ongoing*

Indian Institute of Technology, Madras — Tamil Nadu, India
M.S.-Research, Department of Civil Engineering — *Aug 2021 – June 2023*
CGPA: **9.6/10**

National Institute of Technology, Silchar — Assam, India
B.Tech., Department of Civil Engineering — *Jul 2016 – May 2020*
CGPA: **9.42/10**

Research Projects

Fluid-induced Frictional Ruptures in Fractured Rock — EPFL
Doctoral Research — *Sept 2023 – ongoing*

- The objective of the thesis is to conduct comprehensive theoretical and numerical studies to understand the physics of fluid driven frictional ruptures in discrete fracture networks with a particular focus on fault damage zones.

Physics Informed Neural Networks for Heterogeneous Subsurface Systems — IIT Madras
MS-Research Thesis Project — *Aug 2021 – June 2023*

- Developed a robust framework of Physics Informed Neural Networks (PINNs) that can be used in modelling heterogeneous physical systems with material interfaces.
- This framework is expected to have significant applications in modelling geometrically complex & highly heterogeneous subsurface systems having geological stratification along with naturally occurring faults.

Comparison of Different Dewatering Techniques for Improvement of Soft Soil — NIT Silchar
B.Tech Thesis Project — *Aug 2019 – May 2020*

- Carried out a case study along a stretch of NH-37 in Silchar, India. The objective of the case study was to compare three de-watering methods namely, vacuum, electro-osmosis, and prefabricated vertical drains with respect to the improvement they provide to the bearing capacity of the underlying soft strata.

Stability Analysis of Soil Slopes in Meghalaya, North-Eastern India — NIT Silchar
Independent Research — *Mar - Dec 2019*

- Computed factor of safety and identified critical slip surface of four soil slopes near NH-40, a strategic road corridor in Meghalaya, India with limit equilbrium method.

Research Publications

- **Sarma, A.K.**, Roy S., Annavarapu, C., Roy, P. and Jagannathan, S. (2024), *"Interface PINNs (I-PINNs): A physics-informed neural networks framework for interface problems"*. Computer Methods in Applied Mechanics and Engineering, Volume 429, 117135. Link

- Roy S., Annavarapu, C., Roy, P. and **Sarma, A.K.** (2024), *"Adaptive interface-pinns (adai-pinns): An efficient physics-informed neural networks framework for interface problems"*. Accepted in Communications in Computational Physics. Link

- **Sarma, A.K.**, Annavarapu, C., Roy, P. and Jagannathan, S. (2023), *"Variational interface physics informed neural networks (VI-PINNs) for heterogeneous subsurface systems"*. 57th U.S. Rock Mechanics / Geomechanics Symposium, ARMA-2023-0760. Link

- **Sarma, A. K.**, Choudhury, A., Sarma, K. and Alammyan, A. (2019) *"Stability Assessment of a Soil Slope in Meghalaya, North-Eastern India"*. Lecture Notes in Civil Engineering, Volume 133 LNCE, Pages 753 - 7632021 Indian Geotechnical Conference, IGC 2019, Surat, Dec 19 - Dec 21, 2019, 256869. Link

Conference Presentations & Posters

- **Sarma, A.K.**, Lecampion B. (2024), *"Fluid Injection Induced Seismicity: A Numerical Study of Aseismic Cascade Slip Events in Fault Damage Zones"*, poster presented at EARTHQUAKES: nucleation, triggering, rupture, and relationships to aseismic processes – 4th edition, Cargèse international workshop on earthquakes, 14-18 Oct 2024 Cargèse, Corsica.
- **Sarma, A.K.**, Annavarapu, C., Roy, P. and Jagannathan, S. (2022), *"PINNs for Heterogeneous Materials"*, presented at 15th World Congress on computational Mechanics & 8th Asian Pacific Congress on Computational Mechanics, **WCCM-APCOM 2022**, Yokohama Japan Virtual, July 31st - Aug 05, 2022.

Experience

Teaching Assistant EPFL

Feb 2024 - ongoing

- Conducting weekly exercise sessions and grading assignments and exams for courses "Civil 306: Geotechnical works", and "Civil 423: Computational Geomechanics" instructed by Prof. Brice Lecampion.

Teaching Assistant IIT Madras

Jul 2022 - May 2023

- Conducted weekly MATLAB tutorial sessions and graded assignments in the course "CE 5815: Introduction to Computational Geomechanics", instructed by Dr. Chandrasekhar Annavarapu.

Leadership & Community Service

President, YUVA-Indians EPFL

Jul 2024 - ongoing

- Leading the association YUVA-Indians (Youth United via Action) which is a public, non-profit, non-political, and non-religious association for Indian and Indian-origin students studying or working in Lausanne.

Research Affairs Secretary Executive IIT Madras

Jul 2022 - April 2023

- Addressed queries and concerns of the research scholars' community of IIT Madras by acting as a bridge between administration and scholars.

Executive Committee Member, IGSSSC NIT Silchar

Jun 2019 - May 2020

- Organised various conferences, workshops & field visits as a part of activities conducted by Indian Geotechnical Society Silchar Students' Chapter (IGSSSC).

Scholastic Achievements & Awards

- Recipient of **Swiss Government Excellence Fellowship 2024**.
- Recipient of "Summer Research Fellowship Programme 2019", conducted by Indian Academy of Sciences.
- Recipient of "Chief Minister's Academic Excellence Fellowship" for being in top 10 rankers in State.
- Awardee of "Anundoram Borooah Award (2014)" for academic excellence in secondary examination.
- Secured **2nd** rank in B.Tech in Department of Civil Engineering, NIT Silchar.
- TOEFL Score: **110**. (Test date: September 25, 2022)
- Graduate Aptitude Test in Engineering (GATE 2021) qualified with **99.03** percentile.

Technical Skills

Modeling Techniques: Finite Element Method, Boundary Element Method, Data-Driven Modelling, ML
Programming Languages: Python, MATLAB, C & C++ Programming
Softwares & Tools: OPTUM, PLAXIS, ANSYS, AutoCAD, GeoStudio, LaTeX, MS Office, GitHub etc.